

COURSE SYLLABUS

CSE200: RESEARCH & TEACHING | FALL 2025 | UNIVERSITY OF CALIFORNIA, SANTA CRUZ

COURSE INFORMATION

Location: Humanities Lecture Hall | **Day/Time:** Tuesdays, 5:20 PM – 6:55 PM

Discussion section dates/times are available via MyUCSC, occurrences are tentative.

COURSE DESCRIPTION

This course focuses on introducing graduate students to the practical aspects of **starting graduate-level research** and **equity-focused, effective teaching**. It is completely in-person and is meticulously designed to be an active learning and collaborative experience for both students and the teaching team. The course consists of both lectures — which include presentations, workshops, and guest speakers — and discussion sections — which include collaborative discussions, small group activities, and application of concepts covered in lecture. Students of this course are exposed and introduced to various topics, strategies, and resources in research and teaching and are encouraged and guided to explore them further for their personal and professional development. Students are encouraged to engage in thoughtful discussion and reflection throughout the course and are motivated to apply their skills and findings in following quarters as they begin their research and assume their roles as teaching assistants. Ultimately, this course will focus on providing students the tools they need for **success in graduate school** and in turn developing **stronger and more connected cultures of research and teaching** within the Computer Science and Engineering department at the University of California, Santa Cruz.

TEACHING TEAM



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TBD
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LEARNING OBJECTIVES

RESEARCH

1. **Culture of Research.** Understand the landscape and culture of research done at BSOE and beyond.
 - a. *M.S. students:* Explore and interact with different research groups to select an area and topic for your thesis or project.
 - b. *Ph.D. students:* Locate the positionality of your research in the broader research landscape at UCSC and form connections with faculty and peers within and beyond your lab.
 - c. Understand the importance of conducting research while maintaining personal and professional ethical considerations.
2. **Technical Resources.** Get familiar with technical resources provided by the University and BSOE and why you may need them.
 - a. Use or access the various computing resources available on campus.
 - b. Use or access hardware resources (such as laptops, servers, VMs, computing accounts, and more).
3. **Personal Development.** Practice and make plans to access resources for the necessary technical communication and interpersonal skills for success in graduate school.
 - a. Practice collaborating with your peers and providing feedback.
 - b. Connect with resources to improve your technical writing and presentation skills.
 - c. Connect with resources on personal and professional development.
4. **Academic Literature.** Learn how to explore, analyze, and contribute to scientific literature in the research context.
 - a. Use on-campus tools and resources for exploring and accessing scientific literature.
 - b. Use software tools and resources for exploring, managing, and annotating scientific literature.
 - c. Read, infer, analyze, and communicate thoughts and perspectives about scientific literature.
5. **Research Formulation.** Learn how to synthesize novel research ideas and formulate relevant research questions.
 - a. Learn how to brainstorm, explore, and research new topics and ideas.
 - b. Define the scope and objectives of your planned research.
 - c. Develop research questions that are specific, measurable, achievable, relevant, and time bound.

TEACHING

1. **Teaching Teams.** Reflect on the positionality of graduate student educators between the students and instructors and gain tools for leveraging that positionality effectively.
 - a. Become familiar with your roles and responsibilities interfacing between students and instructors.
 - b. Communicate effectively with instructors to help facilitate a positive teaching environment.
 - c. Use appropriate channels, methods, and resources to address academic integrity and other conflicts and issues.
 - d. Employ ethical considerations as it pertains to the instruction of students, teaching course materials, and interfacing with teaching teams.
2. **Classroom Orchestration.** Gain insight into various aspects of conducting and facilitating discussions and lab sections.
 - a. Communicate effectively and empathetically with students.
 - b. Explore and plan teaching techniques to engage students and encourage active participation.
 - c. Design discussion sections whose learning objectives follow those of the curriculum.
 - d. Learn and utilize conflict recognition and resolution techniques.
 - e. Teach and learn in different contexts, including programming-based sections, discussion sections, lab sections, and office hours.
 - f. Handle different teaching modalities — including online, hybrid, and in-person environments — and acquire strategies to work within them.
3. **Equity-minded Teaching.** Recognize and learn how to take steps to improve equity gaps in the teaching and learning experience through data-driven methods and frameworks.
 - a. Gain awareness of and ability to recognize systemic and situational challenges impeding student learning.
 - b. Gain insight on how to connect with their students and recognize their backgrounds and capital to create a responsive learning environment.
 - c. Gain insight into recognizing student performance gaps and creating effective solutions.
 - d. Foster an inclusive, equitable, and accessible teaching and learning community within the department and the broader University.

COURSE FORMAT & SCHEDULE

This course will consist of lectures, discussions, and hosted sessions with guest speakers. Most of these events will happen during the Tuesday main sessions, with the discussion sections being tentative and reserved for in-depth activities. **NOTE:** Attendance is mandatory for both the weekly main session and the tentative discussion sections (when they happen).

The following schedule is tentative and is subject to change in response to the course needs and circumstances. Below the description of each week, you will also find the correlated learning objectives, in which **R** refers to the Research Learning Objectives and **T** refers to the Teaching Learning Objectives.

Course Introductions & Orientations <u>Week 1:</u> R3	Course Introduction & Syllabus Welcome + Graduate Advising Orientation UAW (Union) Orientation
Introduction to Research <u>Week 2:</u> R1, R3	Introduction & Overview of Research at UCSC <i>Guest Speaker(s) on Ongoing Research</i>
Tools for Exploring & Doing Research <u>Week 3:</u> R2, R4, R5	Exploring Research & Research Hygiene Campus Computing Resources Advanced Research Tools & Library Resources
Reading & Writing Academic Literature <u>Week 4:</u> R3, R4, R5	How to Read an Academic Paper Analyzing Academic Literature <i>Guest Speaker(s) on Ongoing Research</i>
Research Tools & Professional Communication <u>Week 5:</u> R1, R2, R3	Professional Communication in Academic Settings Applying to Grants & Fellowships Career Development & Job Search
Introduction to Teaching <u>Week 6:</u> T1, T3	Introduction to Teaching at UCSC Foundational Teaching Theories
Robust & Effective Teaching Practices <u>Week 7:</u> T3	Analyzing the CSE Context through Student Data Conversations on Equity & Inclusion Inclusive Teaching in STEM Disciplines
Active Learning & Managing Environments <u>Week 8:</u> T2, T3	Active Learning + Strategies Section Organization Classroom & Lab Environments
New Tools in Teaching & Learning <u>Week 9:</u> T2	Generative AI in Teaching & Learning Contexts Academic Integrity
Course Epilogue <u>Week 10:</u>	<i>Social Event(s)</i>

COURSE REQUIREMENTS



There are **no course prerequisites** for this course! Unlike some of the more technical courses you might take in your graduate school career, this course is a personal and professional development course, largely based on the positionality of being a graduate student and its responsibilities in research in teaching. This course is intended for graduate students who are expected to be starting graduate-level research and/or teaching assistantships in the following quarters after taking it.

Accordingly, we encourage you to engage in this course while thinking about how you might apply the concepts, skills, and resources from this course in your future graduate career.

Material Requirements. The only material requirement for this course is **access to a computer that can access the Internet**. If this is unavailable to you, university computers are available through computer labs across campus. **You are encouraged to bring laptops to the sessions to engage with online activities**; phones and tablets are welcome but may present more barriers in engagement. If you need to loan a computer for this course, you can check one out from the [university libraries](#). You will be accessing the Canvas website or app as the main hub for this course, using Google Workspace with your school account for peer collaboration, and accessing web resources and software tools that will be presented to you.

COMMUNICATION

You are always encouraged to communicate with the Graduate Student Instructor (GSI) or Teaching Assistant (TA) if you have any questions or concerns regarding the course. **For quick conversations**, you can talk to the GSI or TA after lecture, as time permits. **For longer conversations**, we invite you to our student hours which are listed in the [Teaching Team](#) section of this syllabus or make an appointment with us via email. Student hours and appointments will normally be accessible in-person and online via Zoom. **For text communication**, we invite you to use Canvas messaging to reach out to the GSI or TA and Canvas forums to ask questions to your peers. You can also use email to contact us; we ask that you include a relevant subject line that mentions the course number, “CSE200”, so that we can easily find your email. We will do our best to give you timely responses during our working hours **Monday to Friday from 9 AM to 5 PM**, as our own schedules permit (we’re students and researchers, too!). Our communication will be more limited outside of these hours and on weekends and holidays. **Class communications (announcements, universal extensions, etc.) will be made on the Canvas site.**



Accommodations. In addition to questions and concerns about the course, we **encourage you to reach out for any accommodations due to personal or academic circumstances or emergencies** that impact your ability to attend sessions, engage in activities, or submit assignments. In these cases, we are willing to work with you to figure out solutions that ensure your success in this course.

STUDENT FEEDBACK



This course was experimentally redesigned for the Fall 2023 quarter by the teaching team was heavily supported by input from faculty members, past student feedback, and the Teaching and Learning Center (TLC). As we continue to develop and refine the course for this year and following years, **we invite you to provide your own feedback and perspective**, so that we can use it to improve your experience and future graduate students' experiences. Throughout the quarter, we will provide opportunities for feedback through **reflection questions**, **exit tickets** (small surveys that will occasionally happen at the end of a session), and **discussions**. You will also be given the opportunity to provide feedback through **SETS** later in the quarter.

YOUR SUCCESS IN THIS COURSE

This course is a *Satisfactory-Unsatisfactory* graded course (also sometimes referred to as a *Pass-No Pass* course). **This does require action on your part. You must request a change in grading option by the published deadline for the quarter (November 30, 2025).** We recommend that you request the change of grade as soon as possible.

We want to ensure your success in this course, and that success depends on consistent attendance, engagement, and practice. Getting a *Satisfactory* or passing grade in this course requires you to satisfy three (3) requirements:

- **Attend lectures and discussions consistently.**
- **Engage fully, respectfully, and honestly in activities and discussions.**
- **Receive a satisfactory grade average across all assignments.**
- **Attend at least 3 colloquium talks (participation submission via form).**

If we notice - or you are worried - that you are at risk of not satisfying these requirements, we can check in with you to figure out how you can get back on track.

ATTENDANCE & ABSENCES

Because this course is a core requirement for your graduate program and consists of practice activities, discussions, and reflections, your attendance to every lecture, discussion, and hosted session is required. Because many of our sessions will involve small group activities, any unexpected absence will be noted. We thank you for making the time and commitment to attending this course.



Absences. We do understand that life happens, and there may be situations when you are unable to come to the sessions. If you must miss any of the sessions, we ask that you **communicate to the teaching team via email about your absence, ahead of time if possible**. We will work with you on a case-by-case basis to ensure that your absence will not impede your ability to do well in this course and provide you with the tools and resources to necessary to do so.

CLASSROOM ENGAGEMENT



This course utilizes many of the active teaching and learning strategies that it teaches, including discussions with your peers and the teaching team, small group activities, collaborative projects, and self-reflections. In order to more deeply connect with and consistently practice the skills and resources you take from this course, **we ask that you engage fully, respectfully, and honestly with the discussions, activities, and reflections.** Good participation

and engagement can help build a community of peers that can carry with you for your coming years of graduate school.

ASSIGNMENTS, SUBMISSIONS, & FEEDBACK

This course encourages you to practice skills, explore topics, and reflect on your progress through a few written assignments over the course. We try to keep these assignments - intended for you to work on *between* sessions - minimal, as we want you to have adequate time and energy to spend on in-class discussions and activities and, of course, your other courses as well. While we want to provide you with ample time and flexibility to develop and submit your work, **assignment deadlines are important for**



time management on part of both the students and the teaching team. Deadlines allow for students to maintain a consistent pace throughout the course and for the teaching team to provide timely and meaningful feedback. Timely and good-quality submission of assignments is also important for in-class discussions and activities that follow and integrate with the assignments. **Assignments can be accessed, submitted, and reviewed on Canvas.**

Grading and Feedback. While the course will be graded as *Satisfactory/Unsatisfactory*, all assignments will receive more fine-grained evaluations via points that collectively inform whether or not you receive credit for the course. As your teaching team, we aim to provide you with encouraging and constructive feedback on each of your submissions as we evaluate your work in order to support your personal and professional development as a graduate student.

Late Submission. To balance timeliness and flexibility, **you are given 4 total late days to use on any assignment submission.** You can use these late days for any assignment without consequence and are applied automatically. After your late days are used up, **late submissions will incur a 20% penalty on the assignment per each day late**, applied after the assignment has been evaluated for its content. Remember that we are always open to providing accommodations in special circumstances.

ATTENDING CSE COLLOQUIUMS

Becoming familiar with state-of-the-art research – and more importantly, the researchers and kinds of discussions that define today’s academic landscape – is a critical aspect in your pursuit of a M.S. or Ph.D. degree. Our department holds a series of **colloquiums**, academic forums and presentations on the current state of research in various disciplines, every quarter. To encourage you to utilize this opportunity to build habits around exploring and attending research discussions, **we are requiring that you attend at least 3 colloquiums during the quarter.**



Confirming Participation via Form. To notify us of your participation, **you will fill out a Google Form that confirms your participation.** This form will be made available on Canvas to you before the first session.

STUDENT & PROFESSIONAL CONDUCT



As a personal and professional development course, this course will involve interfacing with guest speakers, the teaching team, and your peers. In addition to people, you will also be interfacing with resources, materials, and tools. **Conduct yourself with dignity and respect and treat others with the same.** Building a conducive environment for learning and collaboration starts with dignity and respect. As you are graduate students, **you are professionals, so act and converse professionally.** This does not mean that casualness or informality are not allowed, but rather that you consider the topics of discussion or presentation with importance and imperative.

Conducting Dialogue. *This statement has been adapted from the TLC and sourced from the University of Michigan.* **In some of our in-class discussions and dialogues, we will have the opportunity to explore challenging, high-stakes issues and increase our understandings of different perspectives.** Our conversations may not always be easy, and we sometimes will make mistakes in our speaking and our listening. Sometimes we will need patience or courage or imagination or any number of qualities in combination to engage our resources, our peers, and our own ideas and experiences. Always, we will need respect for others. Thus, an important aim of our session interactions will be for us to **increase our facility with the conversations that arise inside issues of graduate research and equitable teaching as we deepen our understandings of multiple perspectives** – and make the most of being in a room with people of many backgrounds, experiences, and positions.

ACADEMIC INTEGRITY

Academic integrity is a commitment to trust, respect, and honor through responsibility and accountability. In this course, you are exposed to numerous sources ranging from guest speakers to scientific literature and you will most likely consult numerous tools including websites, search tools, and generative AI (artificial intelligence). **The use of and reference to all such sources and tools are allowed if these sources and tools are used responsibly** — i.e., appropriately, after ensuring its credibility,



and in a way that supports, not replaces, your work — **and with proper credit** — e.g., citations, captions, or accreditation. You are also exposed to the original thoughts and ideas of your peers and teaching team, and **you are welcome to use or refer to your peers' and your teaching team's original thoughts and ideas if you properly cite and uplift them.** By citing and uplifting all the sources and ideas that bring value to you and your contributions, you help to build and maintain the collaborative, constructive, and inclusive environment that is the foundation for effective research and teaching. You can find more information about academic integrity and academic misconduct at the [*Academic Integrity for Graduate Students*](#) webpage.

IMPORTANT CAMPUS RESOURCES

The following resources can serve you as necessary, both within this course, and throughout your graduate career. If you need a resource that is not listed, please do not hesitate to contact me, and we can do our best to connect you with a resource that serves you.

DISABILITY RESOURCE CENTER (DRC)

The following statement has been provided by the DRC for inclusion in this course syllabus. UC Santa Cruz is committed to creating an academic environment that supports its diverse student body. If you are a student with a disability who requires accommodations to achieve equal access in this course, please affiliate with the DRC. We encourage all students to benefit from learning more about DRC services to contact DRC by phone at (831) 459-2089 or by email at drc@ucsc.edu. For students already affiliated, make sure that you have requested Academic Access Letters, where you intend to use accommodations. You can also request to meet privately with me during my office hours or by appointment, as soon as possible. I would like us to discuss how we can implement your accommodations in this course to ensure your access and full engagement in this course. You learn more about the DRC at their website: <https://drc.ucsc.edu/>.

COUNSELING AND PSYCHOLOGICAL SERVICES (CAPS)

As a graduate student, maintaining your mental health while handling multiple responsibilities can be difficult. If you are in distress, experiencing heightened stress and anxiety, or want to get more support and perspective on something you are going through, CAPS provides a variety of services for your needs - including immediate crisis support, scheduled individual appointments, group counseling, and workshops led by peer advisors. You can find more resources, contact information, and hours at their website: <https://caps.ucsc.edu/index.html>.

CENTER FOR ADVOCACY, RESOURCES, & EMPOWERMENT (CARE)

CARE is a confidential space to discuss issues of dating violence, sexual assault, and stalking. CARE advocates provide support in a variety of ways depending on your needs, such as by supporting you in your decision-making; understanding the complexities that can arise from these issues; providing emotional support and free services; and providing resources and referrals. You can make an appointment and find more resources and contact information at their website: <https://care.ucsc.edu/>.

TITLE IX

The following statement has been provided by the Title IX office. Title IX prohibits gender discrimination, including sexual harassment, domestic and dating violence, sexual assault, and stalking. If you have experienced sexual harassment or sexual violence, you can receive confidential support and advocacy at the Campus Advocacy Resources & Education (CARE) Office by calling (831) 502-2273. In addition, Counseling & Psychological Services (CAPS) can provide confidential, counseling support, (831) 459-2628. You can also report gender discrimination directly to the University's Title IX Office, (831) 459-2462. Reports to law enforcement can be made to UCPD, (831) 459-2231 ext. 1. For emergencies, call 911. Faculty and Teaching Assistants are required under the [UC Policy on Sexual Violence and Sexual Harassment](#) to inform the Title IX Office should they become aware that you or any other student has experienced sexual violence or sexual harassment. You can learn more about Title IX, your rights, and your protections at their website: <https://titleix.ucsc.edu/index.html>.

ACKNOWLEDGMENTS



This course has been developed for this quarter by a team of graduate students— Harikrishna Kuttivelil (CSE Graduate Pedagogy Fellow 2022), Andrea David (CSE Graduate Pedagogy Fellow 2023), Lakshmi Krishnaswamy (CSE Graduate Pedagogy Fellow 2024), and Ian Terry (CSE Graduate Pedagogy Fellow 2025) — following the Teaching and Learning Center's (TLC) Graduate Pedagogy Fellowship Programs of 2022-2023, and other fellowships and programs provided by the TLC. **We appreciate the skills, resources, and materials that the TLC has provided us, along with their counsel and feedback that has guided the development of the course.** To learn more about the TLC, you can visit them at the [TLC website](#).

We designed this course with input from our advisor, Dr. Katia Obraczka, who we thank for her guidance. This course was adapted from the previous *CSE200: Research and Teaching* course taught by previous faculty instructors, including Dr. Katia Obraczka, Dr. Chen Qian, Dr. Roberto Manduchi, and more. **We thank these faculty, and more who participated and gave input to this course, for their valuable insight. We also thank current and former graduate students, from the CSE department and others, whose perspectives and feedback have also shaped this redesigned course.**

LAND ACKNOWLEDGMENT

The land on which we gather is the unceded territory of the Awaswas-speaking Uypi Tribe. The Amah Mutsun Tribal Band, comprised of the descendants of indigenous people taken to missions Santa Cruz and San Juan Bautista during Spanish colonization of the Central Coast, is today working hard to restore traditional stewardship practices on these lands and heal from historical trauma.

